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Intelligent Email Spam Detection: A Machine Learning-Based Approach

Vinod Jain
Department of Computer Engineering and Applications
GLA University
Mathura, India
vinod.jain@gla.ac.in
(ORCID-0000-0003-0260-7319)

***Abstract*—Email spam emails frequently contain dangerous material like viruses, phishing attempts, and fraudulent schemes. Email spam detection is an essential component of cybersecurity. Machine learning models are required to increase accuracy and efficiency because traditional rule-based filtering strategies are insufficient to handle developing spam techniques. This study investigates how well five popular machine learning algorithms—Random Forest, Logistic Regression, Support Vector Machine (SVM), Decision Tree, and Multinomial Naïve Bayes (MultinomialNB) to detect and categorize spam emails. Different methods are used by each of these models to examine email content and detect spam. MultinomialNB is a probabilistic classifier founded on the Bayes theorem, excels at text classification problems. By merging several decision trees, Random Forest, an ensemble learning technique, improves generalization and decreases overfitting while increasing prediction accuracy. A linear model for binary classification called logistic regression uses learned weights for several features to calculate the likelihood that an email is spam. SVM, a potent classification method, finds the best hyperplane in a high-dimensional space to distinguish between spam and non-spam emails. A rule-based approach called a Decision Tree uses a hierarchical structure of decision nodes to classify emails, making it simple to use and analyze. A dataset of tagged spam and non-spam emails is used to assess the performance of these models. To ensure a thorough evaluation of the model, the dataset is divided into training and testing sets. To compare the efficacy of the models, evaluation criteria accuracy is employed. While all models perform well, the results show that SVM offer superior accuracy and achieved 98.45% accuracy. The results emphasize how crucial it is to choose the right machine learning models depending on the properties of the dataset, available computing power, and intended performance indicators.**

Keywords—Machine Learning; Email Spam Detection; Logistic Regression; Random Forest; Decision Tree; Naïve bayes

I. INTRODUCTION

As the main channel for both personal and professional contact, email communication is a crucial component of contemporary digital interactions. However, as email usage has increased, spam messages have become much more common, which has resulted in a number of cybersecurity risks such phishing attempts, fraudulent schemes, virus dissemination, and intrusive ads. Spam emails are dangerous because they jeopardize critical data and financial security in addition to using up expensive storage and bandwidth. Conventional techniques for detecting spam, like blacklisting, rule-based filtering, and keyword-based filtering, have been extensively employed. However, because spam messages are always changing, these methods are frequently useless. Spammers always adapt their tactics to get over conventional filters, thus using more advanced methods is crucial. Email spam, often known as junk mail, is made up of unwanted messages sent in large quantities, usually with the intention of phishing, commercial advertising, or malevolent purpose. Phishing emails, which are intended to trick recipients into disclosing private information, malware emails that contain malicious attachments or links, advertising emails that promote unsolicited goods or services, and scam emails that promise financial rewards like lottery winnings or investment opportunities are all examples of spam emails.

Machine learning provides a reliable method for spam detection. In contrast to rule-based approaches, machine learning algorithms enhance categorization accuracy by constantly adjusting to emerging spam trends. Collecting labeled datasets of spam and non-spam (ham) emails is the process of data collection. To distinguish spam from authentic emails, feature extraction finds pertinent characteristics including word frequency, sender details, email structure, and embedded links. While Artificial Neural Networks (ANN) identify intricate patterns in spam messages, Decision Trees and Random Forests use several decision rules to classify emails. For better spam identification, sophisticated deep learning methods such as RNN and LSTM examine the sequential structure of email text.

Higher accuracy than traditional rule-based filters, flexibility in responding to changing spam trends, automation that minimizes human interaction, and scalability to effectively manage enormous email volumes are just a few benefits of machine learning-based spam detection. To improve the precision and effectiveness of spam detection systems, machine learning models must be developed and improved as spam tactics continue to change.

A crucial component of cybersecurity is email spam detection, which shields consumers against phishing, malware, and fraudulent activity. Spam detection has been greatly enhanced by developments in artificial intelligence and machine learning, guaranteeing safer and more effective email communication. As technology advances and research continues, future spam detection systems will become more dependable and effective. The parts that follow will look at

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several machine learning techniques, dataset issues, model evaluation, and future advancements in email spam detection.

II. Related work

One of the biggest problems in cybersecurity has been detecting email spam, which calls for sophisticated categorization techniques. Recent studies have investigated a variety of deep learning (DL) and machine learning (ML) approaches. This review of the literature highlights the methods and conclusions of well-known research that aid in the identification of email spam. A thorough analysis of machine learning techniques used in email spam classification was carried out by M. Raza et al. [1]. The authors emphasized that spam detection rates were increased using hybrid models that combined several machine learning algorithms. Deep learning approaches for email spam detection were investigated by K. Debnath and N. Kar [2]. According to their findings, deep learning models—in particular, CNN-RNN hybrids—performed better than conventional machine learning techniques.

For spam identification, J. Mythili et al. [3] suggested an improved Temporal Naïve Bayes classifier. By using temporal information, their program was able to identify trends in spam emails over time. The findings demonstrated a notable decrease in false positives, enhancing the effectiveness of email filtering. P. Thakur et al. [4], according to their research, ensemble models—Random Forest in particular—achieved the best accuracy in identifying spam emails. A hybrid deep learning model is used by S. E. Rahman and S. Ullah [5]. Their method outperformed stand-alone ML models in spam identification by effectively capturing contextual and sequential patterns in emails.

For business email solutions, R. Wongwatkit et al. [6, 7] presented an Intelligent Blacklist Security System. Their method improved organizational email security by identifying and blocking spam emails using machine learning (ML)-based blacklisting techniques. A hybrid supervised learning strategy that integrated ML classifiers and Natural Language Processing (NLP) was presented by N. U. Saquib et al. [8]. With encouraging outcomes, the authors highlighted the function of feature engineering in raising classification accuracy. In order to detect spam, L. Zou [9] examined both count-based and inference-based NLP models. They concluded that inference-based models provide improved accuracy in spam classification after comparing deep learning-based transformers with conventional TF-IDF.

DistilBERT, a transformer-based model, was used by V. I. Del Rosario et al. [10] to classify emails that contained spam. According to their findings, transformer models provided better spam identification with less computational overhead than traditional ML strategies. Several machine learning techniques for automatic email spam detection were investigated by A. Saini et al. [11]. Their study demonstrated that integrating several classifiers, including SVM and Naïve Bayes, produced a detection system that was more reliable. In their comparative analysis of machine learning algorithms, R. P. Cota and D. Zinca [12] assessed how well each algorithm performed across a range of datasets. Their research confirmed that the best spam detection accuracy was attained by ensemble models, especially boosting-based methods.

Using Natural Language Processing (NLP), G. V. Sesha Sai Krishna Vineeth et al. suggested a revolutionary method for identifying spam emails. Their research focuses on syntactic and semantic text analysis, using NLP methods like lemmatization and tokenization to improve the accuracy of spam classification. The experimental findings demonstrate how NLP-driven methods can be used to improve on conventional spam filtering systems. A Kernel Extreme Learning Machine (KELM) method was presented by S. Priya and R. Annie Uthra to address concept drift in spam identification. Reduced classifier performance results from concept drift, which happens when the properties of email spam change over time. Their research shows that KELM outperforms traditional machine learning models in its ability to adjust to changing spam trends.

In their thorough analysis of optimization methods for spam detection, E. H. Tusher et al. covered issues such unbalanced datasets, hostile attacks, and the requirement for real-time classification. Their research highlights how crucial it is to incorporate hybrid machine learning techniques for better spam detection results.

S. T. Singh et al. investigated a number of Python-based ML and DL algorithms for email spam identification. According to the study's findings, deep learning models perform better in feature extraction and classification tasks than conventional machine learning techniques. In order to improve spam classification performance, S. G. et al. investigated sophisticated machine learning algorithms that integrated feature engineering approaches. Their research assesses how feature selection and representation learning affect spam detection and shows that hybrid methods greatly lower false positive rates. In 2023, D. Yadav and E. B. Kaur reviewed several machine learning models for spam identification and compared how well they performed on various datasets. According to their research, models that have been trained on a variety of datasets exhibit improved generalization and classification accuracy. They also stress how crucial feature selection and hyper parameter adjustment are to ML model optimization. In order to increase the accuracy of spam identification, C. Bansal and B. Sidhu suggested a hybrid machine learning technique that incorporates several classifiers.

Using DistilBERT, a transformer-based NLP model, D. A. Padilla et al. (2024) presented a distributed training method for email spam categorization. Their research demonstrates how well deep learning captures contextual meaning, greatly improving spam detection capabilities. CNN and LSTM networks were used by C. Sai et al. to develop deep learning models for spam classification. Their study emphasizes how crucial deep feature extraction methods are to enhancing classifier robustness. P. N. and P. Jayarekha investigated effective machine learning techniques for classifying spam emails. Their research shows how feature engineering methods. In order to detect English spam emails, S. Kaddoura et al. investigated deep learning algorithms. Their study compares various neural network designs and finds that, when it comes to

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spam detection tasks, LSTM and Bidirectional LSTM (BiLSTM) perform better than conventional ML classifiers. In order to improve email security, Prathyakshini and J. Shetty introduced the Spam Sentinel architecture, which incorporates machine learning techniques. Their work offers insights into potential future research topics by discussing the incorporation of AI-driven models in practical spam filtering systems.

The articles under consideration demonstrate the developments in ML and DL-based spam detection methods. Deep learning architectures, hybrid strategies, and NLP-based techniques all show notable gains in classification resilience and accuracy. To properly handle changing spam patterns, future research should concentrate on real-time classification, adaptive learning models, and adversarial attack mitigation. There are encouraging avenues for improving spam detection effectiveness through the combination of transformer-based models like DistilBERT and distributed training.

III. PROPOSED WORK

A new area of research that has gained greater interest recently is estimating the layout of an indoor space. Identifying incoming emails as "spam" or "non-spam" (ham) is the task of email spam detection. Data gathering, pre-processing, feature extraction, model selection, evaluation, and performance comparison are the main processes in this methodology. The goal is to identify the best algorithm for differentiating between spam and authentic emails by evaluating how well they perform on particular measures like accuracy, precision. Information Gathering - Any spam detection system's initial step is to gather a dataset of labeled emails, each of which is classified as "spam" or "ham." For this purpose, there are a number of publicly accessible datasets that offer a vast collection of both spam and non-spam emails, such as the Enron or Spam Assassin datasets. A training set for training the models and a testing set for assessing their performance should be separated out of the dataset. To prevent biased predictions and guarantee that the model performs effectively when applied to both spam and authentic emails, a balanced dataset is necessary.

Preprocessing Data - Preprocessing the data comes next once the dataset has been gathered. Preprocessing usually consists of the following steps: Text Cleaning: HTML tags, special characters, and formatting that might not be helpful for classification are frequently seen in emails. To guarantee that the models concentrate on the written content, these superfluous elements must be eliminated. Lowercasing: To guarantee case insensitivity, all characters are changed to lowercase as part of the text normalization process. Tokenization: This process divides the email text into smaller chunks, known as tokens. This stage makes processing text data easier and is necessary for the feature extraction that follows. Stop word Removal: During preprocessing, common words like "the," "is," and "in," which don't add much to the meaning of the material, are eliminated. Lemmatization improves model efficiency by bringing disparate word variants together. For instance, "running" turns into "run."

Extraction of Features - Following preprocessing, feature extraction is the next stage. In order for the algorithms to process the data efficiently, text data is usually encoded as vectors of word frequencies. Words in a Bag (BoW): This method uses a vector of word frequencies to represent each email. Every distinct word in the corpus creates a feature, and each feature's value indicates how frequently the term occurs in the email. TF-IDF assist in emphasizing key terms in a particular email by their occurrence. N-grams: These might help catch context that individual words would miss by taking into account sequences of n neighboring words rather than individual words. Word Embedding's: These techniques are especially helpful for capturing context in lengthy or intricate emails. Five distinct machine learning algorithms are selected for email spam detection in this methodology. Figure 1 is showing the proposed model for Email spam detection using five ML models.

MultinomialNB, or Multinomial Naive Bayes Because it is probabilistic, the Multinomial Naive Bayes classifier works very well for spam identification. It considers that features (words) in an email are independent of one another and performs best with discrete data, such word counts. The algorithm is computationally efficient and appropriate for handling big datasets since it determines the likelihood of an email being spam or non-spam based on the observed words in the email. The primary presumption is that, given the class label (spam or ham), the characteristics (words) are conditionally independent.

A random subset of the characteristics and data is used to train each tree in the forest, increasing the model's variety and enhancing its capacity for generalization. Random Forest is an excellent option for spam detection since it can manage noisy datasets and intricate feature interactions. The Logistic Regression Model - A statistical model called logistic regression models the likelihood of a binary event, in this case, whether an email is spam or ham, using a logistic function. For linearly separable data, Logistic Regression can produce good results despite its simplicity. Each characteristic (such as word existence or frequency) is taken into account by the model for spam detection, and it is given a weight that indicates how much it contributes to the classification. The email is classified as either spam or ham by using Logistic Regression, which applies a threshold to the output likelihood.

SVM can handle both linear and non-linear classification tasks and is especially useful for complicated datasets with high-dimensional features. SVM can offer excellent accuracy in the context of spam detection, particularly when the dataset has a lot of features.

Decision Tree- A decision tree is a supervised learning technique that divides the dataset into subgroups based on feature values in order to iteratively construct a tree structure.

Because decision trees may be interpreted, they are simple to comprehend and illustrate. If not adjusted correctly, they may experience overfitting. Pruning methods that establish a minimum number of samples per leaf or a maximum depth can help alleviate this problem.

The training dataset is used to train each of the chosen models. The link between the characteristics and the target label (ham or spam) is taught to the model during training. In order to reduce classification error, model parameters (such as

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weights in logistic regression or tree depth in decision trees) are adjusted during the training phase. During training, cross-validation can be used to avoid overfitting and guarantee that the model works well with unknown data.

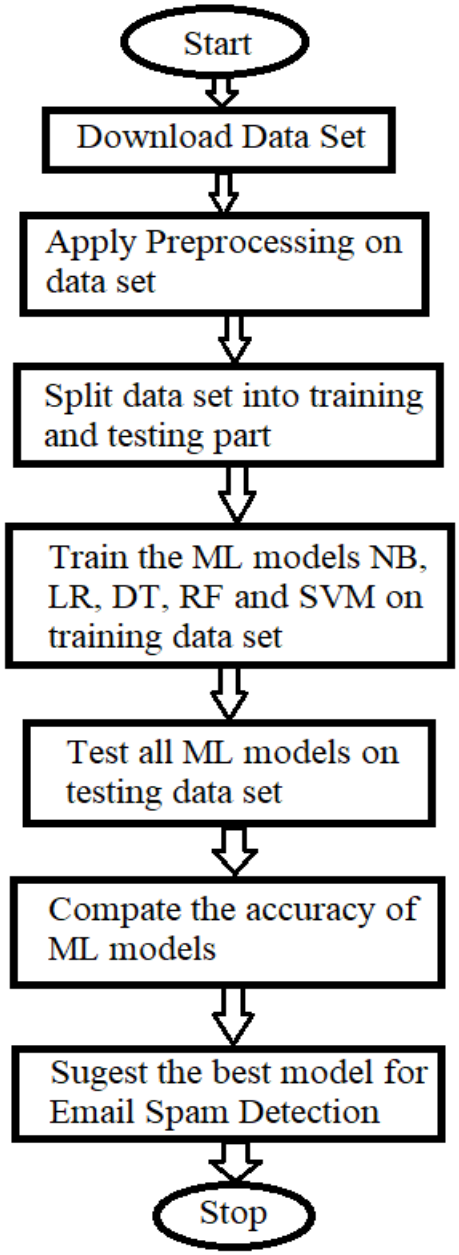


Fig. 1. Architecture of the proposed work

Each model's performance is assessed on the testing dataset following training. The following are typical evaluation measures for categorization tasks: Accuracy: The proportion of emails that are correctly categorized as spam or ham. Precision: The percentage of all anticipated spam emails that are true positives, or successfully recognized spam. Remember: The percentage of real spam emails that are true positives.

The proposed work is implemented using python programming language. The libraries of ML models defined in python language are used for implementation. Anaconda tool is used to run python and its libraries.

IV. RESULTS AND ANALYSIS

Comparing the five models' performances using the evaluation measures is the last stage. The optimal model for spam detection is determined by evaluating its accuracy. To enhance the model's performance, hyper parameter tuning can be carried out if required utilizing methods like grid search or random search. Using the learnt patterns, the chosen model will identify fresh incoming emails as either spam or ham in real-time spam detection. Table 1 is showing the accuracy achieved by five models. It is observed that all the five models are performing well for email classification. Accuracy of more than 95% is achieved by all the five models. It is observed that SVM model is performing best for Email spam detection with 99.9 training accuracy and 98.4 testing accuracy. Figure 2 shows a comparison on the training and testing accuracy of five ML models.

The performance of SVM model is better because this model is robust to outliers, avoids local minima and it is effective for binary classification. All algorithm performs better for training than testing. It shows the problem of overfitting. Future work can be done to resolve this issue.

TABLE I. COMPARATIVE ANALYSIS OF THE EXISTING AND PROPOSED WORKS

Sr. No.	Model	Training Accuracy	Testing Accuracy
1	Naïve Bayes	0.9929	0.9816
2	Random Forest	1.0000	0.9768
3	Logistic Regression	0.9973	0.9807
4	Support Vector Machine	0.9998	0.9845
5	Decision Tree	1.0000	0.9739

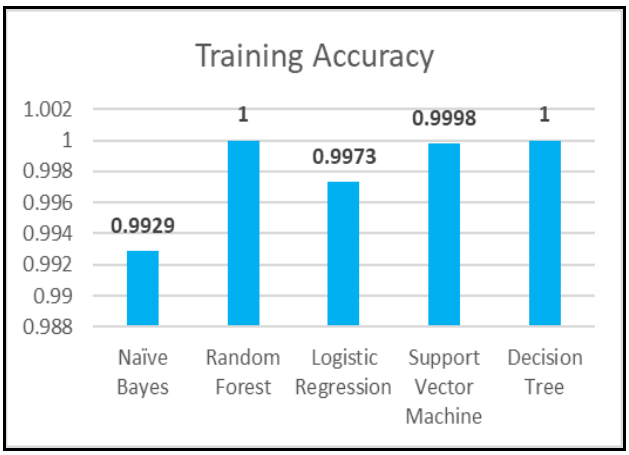


Fig. 2. Performance comparison of ML models based on their training accuracy

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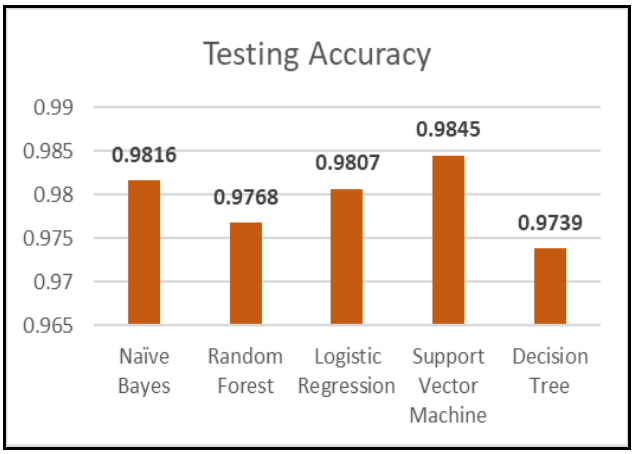


Fig. 3. Performance comparison of ML models based on their testing accuracy

V. Conclusion and Future Scope

The proposed approach uses a number of machine learning techniques to create an efficient spam detection system. We hope to create a reliable system that can correctly identify emails by preprocessing the data, extracting pertinent characteristics, and training and assessing models. The most effective algorithm will be chosen based on performance comparison and implemented in actual email filtering systems to shield users from unsolicited spam messages. SVM is found best among all five ML models with 98.4% prediction accuracy. In future, further work can calculate other performance metrics such as precision, recall, reliability etc. The work mainly calculate the training and testing accuracy of five ML models. The performance of these models can be further improved using cross validation etc. Further the performance can be compared with other ML models applied by other researchers on the same data set.

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